

**a. Graph 1: Percentage of waste volume per region in Jakarta**

**b. Graph 2: The average daily volume of waste in Jakarta**

\*The text below is the description for graph 1 and graph 2

The first chart shows the percentage of total waste volume by administrative region in Jakarta over one month, while the second chart displays the average daily waste volume per region in cubic meters (m<sup>3</sup>).

Interestingly, both charts show a very similar distribution pattern, with the highest order consistently being: West Jakarta, Central Jakarta, South Jakarta, East Jakarta, and North Jakarta. The only difference is in the positions of the last two regions, which are East Jakarta and North Jakarta.

West Jakarta ranks the highest in both total and average daily waste volume. This area is known as one of the most densely populated and economically active regions, with crowded residential areas like Tambora, Cengkareng, and Kalideres, as well as economic centers such as Pasar Kopro and Jembatan Lima.

Central Jakarta follows as a major contributor, even though it has a smaller area. It serves as the administrative and economic center of Jakarta. Areas like Tanah Abang, which is the largest textile market in South-east Asia, along with office zones in Menteng and Gambir, lead to high daily mobility, which contributes to the high waste volume.

South Jakarta, which ranks third, has a mix of upper-middle-class residential areas as well as shopping and office centers, such as those in Kebayoran Baru and Setiabudi. This explains why its waste volume is relatively high, though not as high as in West and Central Jakarta.

On the other hand, East Jakarta and North Jakarta record lower waste volumes. In East Jakarta, although it is the largest in area among all region, it is mostly made up of medium-density residential zones and many open spaces, such as Buperta Cibubur and Taman Mini Indonesia Indah.

Meanwhile, North Jakarta consists largely of ports, industrial zones, and reclaimed land. Areas like Tanjung Priok and Sunter are strongly industrial and logistics-oriented, and not all of the waste from these activities is recorded as domestic waste.

**c. Graph 3: Top 5 Sub-Districts with the Highest Waste Volume in DKI Jakarta**

The next chart shows the five sub-districts with the highest total waste volume during the one-month period. All five of these sub-districts are located in West Jakarta, namely: Taman Sari, Grogol Petamburan, Kembangan, Gambir, Palmerah.

Taman Sari is a densely populated residential zone and is home to many large traditional markets, such as Pasar Pagi Mangga Besar and the Glodok area. The high concentration of small businesses and retail activities generates a large amount of daily waste, especially from packaging, food waste, and other organic materials.

Grogol Petamburan is known as a densely populated area with a mix of residential zones, educational institutions, and shopping centers. It is home to major universities such as Tarumanagara University and Trisakti University, as well as large malls like Central Park, Neo Soho, and Taman Anggrek Mall. The high daily activity from students, mall visitors, and local residents contributes to the large amount of domestic and commercial waste produced.

Kembangan has experienced rapid development over the past decade as a modern residential and office area, such as the Puri Indah area. Although it is not as densely populated as Taman Sari, its waste volume is high due to increasing construction activities and household waste from upper-middle-class housing, as well as many restaurants and small business hubs.

Gambir, although small in area, is the national administrative center. Central government offices such as the Ministry of Home Affairs, the Audit Board (BPK), and even the Presidential Palace are located in this sub-district. Daily administrative activities and high human traffic in the area lead to a large amount of institutional and daily consumption waste.

Palmerah, which is a mixed area of residential, office, and media zones, also records a high waste volume. This area includes major media offices such as Kompas Gramedia and traditional markets like Pasar Palmerah.

#### **d. Graph 4: Daily Waste Volume Trend**

Next is the chart showing the trend of daily waste volume over one month. We can see that the waste volume peaks on the 9th, 18th, and 29th. However, the data does not state the specific month. To understand the context of these spikes, we compared the dates with the national calendar in 2024.

Assuming the data is from June 2024, some interesting findings emerge:

- June 9, 2024, falls on a Sunday, when public activity tends to increase due to family outings, recreation, or weekend markets.
- June 18, 2024, is a public holiday (Collective Leave for Eid al-Adha 1445 H), which is usually followed by animal sacrifices and communal meals. This likely leads to a large amount of organic waste.
- June 29, 2024, falls on a Saturday, closing out a long weekend. Increased activity during the weekend can cause waste buildup that may carry over into the following Monday.

A strategic step to address this is adjusting waste collection schedules and capacity ahead of weekends or holidays, to prevent excessive waste buildup.

#### **e. Graph 5: Relation between area and waste volume**

Lastly, we look at the scatter plot chart that compares the length of administrative areas with the average daily waste volume. The analysis shows a Pearson correlation of -0.0876, which is close to zero. This indicates that there is no significant linear relationship between the size (length) of an area and the amount of waste it produces. In other words, larger areas do not necessarily produce more waste, and smaller areas do not always produce less.

And why is that? Waste volume is more influenced by social and demographic characteristics, such as Population size and density, Land use types like residential, industrial, or markets, and Daily community activities

Central Jakarta, although small in area, has the second highest daily waste volume. This shows that waste volume is influenced by the function and intensity of human activities, not just by the size of the area. Meanwhile, North Jakarta, despite being larger, produces less waste due to the dominance of ports, industrial zones, empty lands, and coastal areas, with fewer residential zones.

In conclusion, physical indicators like area size should not be the main predictor in waste management. A more holistic approach is needed, taking into account social, economic, and behavioral aspects of the community.

### **Conclusion :**

From this analysis, it is shown that the volume of waste in Jakarta is significantly influenced by the density of activities and the characteristics of the area, rather than just the area size. Areas such as West Jakarta and Central Jakarta stand out in terms of waste volume due to the high intensity of economic, commercial, and social activities.

As a first step, the enhancement of waste management should be focused on areas with the highest volume, especially by increasing transportation efficiency, expanding sorting facilities at the source, and strengthening public education regarding waste reduction and management.

Furthermore, the integration of population data, economic activities, and types of waste is needed to support more targeted policymaking.

### **area of each region in Jakarta :**

<https://www.jakarta.go.id/tentang-jakarta#:~:text=Kota%20Administrasi%20Jakarta%20Pusat%20dengan%20luas%2048%2C13,Kepulauan%20Seribu%20dengan%20luas%20area%208%2C70%20km%C2%B2.&text=Penduduk%20DKI%20Jakarta%20tercatat%20sebanyak%2010.644.776%20jiwa%20per%20tahun%202021>.